

CLINICAL INVESTIGATION

Involved Neck Only Versus Mucosal Radiation Therapy for Head and Neck Squamous Cell Cancer of Unknown Primary (HNSCCUP): A National Retrospective Multicenter Cohort Study



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Purpose: Target volumes for irradiation remain ill-defined for head and neck squamous cell cancer of unknown primary (HNSCCUP). The aim of this study was to compare 2 commonly used radiation therapy strategies for patients diagnosed with HNSCCUP: ipsilateral or involved neck only (INO) versus bilateral neck and/or mucosal (MUC) radiation therapy, evaluating disease-related outcomes and enteral feeding rates.

Methods and Materials: This was a retrospective, observational, multicenter cohort study. Patients diagnosed with unilateral HNSCCUP between 2015 and 2023 who underwent radical (chemo) radiation therapy were eligible for analysis. All patients underwent 18F-Fluorodeoxyglucose (FDG) Positron Emission Tomography - Computed Tomography (PET-CT). HNSCCUP was a diagnosis of exclusion made on the basis of negative investigations to detect a primary site. Patient and tumor characteristics, treatment details, toxicities, and disease control were recorded and compared between the 2 radiation therapy strategies.

Results: One hundred ninety-five patients were eligible for analysis, 66% had human papillomavirus-associated disease. Seventy-three patients received INO (37%) and 122 patients received MUC radiation therapy (63%). The median duration of follow-up was 58 months (IQR, 42-72 months). The rate of primary site emergence was 2.7% in the INO and 0.8% in the MUC cohorts, $P = .56$. Five-year overall survival was 80% (95% CI, 70%-90%) for INO and 82% (95% CI, 75%-90%) for MUC radiation therapy, $P = .74$. Those undergoing INO radiation therapy were more likely to die from HNSCCUP (17.8% vs 5.7%), and those receiving MUC were more likely to die from a non-HNSCCUP cause (15.6% vs 4.1%). The need for enteral feeding ≥ 12 months from radiation therapy was 5.7% for MUC versus 0% for INO ($P = .046$).

Conclusions: This is the largest series to date of patients with unilateral HNSCCUP treated radically with radiation therapy in the human papillomavirus and FDG PET-CT era. Acceptably low rates of primary site emergence, lower toxicity, and no difference in overall survival when compared with prophylactic MUC radiation therapy suggest that INO radiation therapy in HNSCCUP may be a feasible alternative in contemporary practice. © 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>)

Introduction

Unknown primary head and neck squamous cell cancer (HNSCCUP) is an unusual entity, and management remains controversial. Although advances in diagnostic techniques mean that a primary site is now more likely to be detected,¹⁻⁴ a minority of patients continue to present with a true occult primary. Management is guided by retrospective studies, which are often limited by small numbers, heterogeneous diagnostic techniques, and treatment. The historical literature is of little relevance to contemporary practice, in part because of advances in diagnostic and treatment technologies, but also due to the changing epidemiology of the disease, with an increasing proportion driven by human papillomavirus (HPV).⁵⁻⁸

Outcomes in HPV-positive HNSCCUP are generally excellent and similar to HPV-positive primary-detected oropharyngeal disease.⁹⁻¹² Therefore, optimizing treatment approaches to minimize late treatment effects without compromising high cure rates is crucial.

Opinion remains divided in the head and neck (H&N) oncology community on the optimal radiotherapeutic approach to HNSCCUP. Although radiation to candidate mucosal (MUC) sites as well as the neck has long been considered standard practice, the necessity of this has been questioned with guidelines now suggesting that either strategy is acceptable.¹³⁻¹⁵ A prospective randomized study, European Organisation for Research and Treatment of Cancer (EORTC) 24001-22005 (NCT00047125), was set up over 20 years ago, comparing bilateral neck and MUC irradiation versus ipsilateral neck irradiation alone, but unfortunately failed to recruit. No further attempts have been made to

generate multicenter, randomized evidence, and it seems unlikely that this will ever be available to guide future radiation therapy practice.

Real-world data provide valuable information beyond the confines of traditional clinical trials and may allow comparison of these approaches.¹⁶

The aim of this study was to compare 2 commonly used radiation therapy strategies for patients diagnosed with HNSCCUP in contemporary practice: ipsilateral or involved neck only (INO) radiation therapy versus bilateral neck and/or MUC irradiation, evaluating disease-related outcomes and enteral feeding rates.

Methods and Materials

Study design

This was a retrospective, observational, multicenter cohort study conducted across 18 UK cancer centers by the NOTCH – National Oncology Trainees Collaborative for Healthcare Research, UK group.¹⁷ Because of the rarity of the disease and good prognosis resulting in a low event rate, a multicenter approach was needed to generate robust analyses that can inform clinical practice.

The study was conducted and is reported according to the Strengthening the Reporting of Observational studies in Epidemiology guidelines.¹⁸

The primary objective of the study was to compare disease-related outcomes between the cohorts, including overall survival (OS), emergence of primary site, and pattern of any other disease relapse, such as contralateral (CL) neck

recurrence. Secondary objectives included a comparison of dosimetry to organs at risk (OARs) and feeding tube requirements (acute and long term) between the cohorts.

Ethical considerations

As this was a retrospective observational study evaluating standard of care treatment, ethical approval was not required. Where appropriate, the NHS England national opt-out service was applied to remove any patients from the study who had previously indicated they did not wish their data to be used for such studies. This option does not currently exist in NHS Scotland or NHS Wales.

The lead center obtained local Caldicott guardian approval to conduct the study and to handle pseudoanonymized data from other sites. To enroll in the study, participating centers were required to gain local Caldicott approval to retrieve and curate relevant data before pseudoanonymized submission to the lead center. Caldicott review ensures that standardly collected data are used legally, ethically and appropriately, and that confidentiality is maintained.

A data sharing agreement between NOTCH and the UK Ear, Nose and Throat (ENT) trainee network, ENT INTEGRATE, was implemented for reasons described below.

Participants

Patients diagnosed with HNSCCUP between January 01, 2015, and August 01, 2023, who underwent radical (chemo) radiation therapy were eligible for analysis. All patients underwent an FDG PET-CT as part of their diagnostic workup. HNSCCUP was a diagnosis of exclusion made on the basis of negative FDG PET-CT, cross-sectional imaging, examination under anaesthetic (EUA), and either imaging directed or blind biopsies.

Inclusion criteria:

- Patients with unilateral cervical node HNSCCUP
- Patients who received radical (chemo) radiation therapy (either adjuvant following neck dissection or as primary treatment)

Exclusion criteria:

- Patients with bilateral HNSCCUP
- Patients who underwent surgery only
- Patients who received palliative treatment [either radiation therapy or systemic anticancer treatment (SACT)] only

Patients were screened and identified from a database established by ENT INTEGRATE. The ENT trainee network had recently published a multicenter study describing the diagnostic workup and surgical management of patients with suspected HNSCCUP across the UK.¹⁹ The dataset was

shared with the NOTCH group. Additional centers that wished to participate but had not contributed to the ENT INTEGRATE study were able to screen and identify patients via locally held tumor board datasets, crosschecked with electronic radiation therapy systems, for the same time period.

Data, including patient demographics, treatment details, toxicities, and disease control, were extracted from either the ENT INTEGRATE database or case records. Staging was categorized as per the American Joint Committee on Cancer 7th edition. Tumor HPV status was assessed using p16 immunohistochemistry (IHC) status and/or HPV- in-situ hybridisation (ISH), as per local unit practice. A positive result from either was regarded as HPV-associated disease throughout this work.

Treatment details

All patients underwent radiation therapy with intensity modulated radiation therapy. A variety of dose and fractionations schedules were utilized: 55 Gy in 20#; 60-66 Gy in 30-33#, and 70 Gy in 35# depending on institutional protocols. Although 55 Gy in 20# is an uncommon regimen internationally, it is widely accepted in the UK as a radical dose schedule and indeed was recommended by American Society for Radiation Oncology (ASTRO) - European Society for Radiotherapy and Oncology. (ESTRO) consensus statement as an alternative to conventional fractionation during the COVID-19 pandemic, which is included in this study time period.²⁰⁻²³

Target volume delineation was categorized into 2 approaches. One entailed treating the INO with a radical dose to involved nodal levels and an elective dose to uninvolved nodal levels in the ipsilateral neck, with no intentional MUC irradiation. The second approach was to treat the involved neck as described as well as potential MUC surfaces and/or the CL neck to a prophylactic dose of radiation therapy (MUC). The decision to employ INO or MUC, and the extent of MUC irradiation in the MUC group, was at the discretion of the treating clinician/center as per local practice. Each center was provided with written descriptions of the INO and MUC categories as well as examples of radiation therapy plans of each to aid classification. The lead center reviewed each case, including radiation therapy plan where necessary, to ensure categorization was consistent. Target delineation in the neck, regardless of INO/MUC strategy, was based on international guidance with the superior border of neck target volume, the inferior aspect of the transverse process of C1.^{24,25}

Concurrent chemotherapy was with cisplatin, carboplatin, or cetuximab, using standard doses and scheduling. Standard OARs were delineated.²⁶

Toxicity

Given the retrospective nature of the work, toxicity evaluation was limited to a feeding tube being inserted in the acute,

treatment phase and being in situ at 12 months or later, from treatment. Doses to OARs were also collected and reported. Relationships between radiation dose to particular OARs and important toxicity endpoints are already well established.^{27,28} An established validated normal tissue complication probability (NTCP) model for EORTC QLQ-H&N35 moderate-to-severe (defined as grade 2+) xerostomia at 6 months was used to show differences between INO and MUC, taking into account baseline status (NTCP values calculated assuming both “no” baseline xerostomia—0 and “a bit” of baseline xerostomia—1, for all patients).

Disease control—primary site emergence

Primary emergence was defined as the new diagnosis of a MUC SCC in an area that could reasonably be the original primary site of disease, eg, in the oropharynx. This required central review of anonymized clinical letters, histology, and radiology reports to differentiate between a new, second primary versus disease consistent with a primary site emergence from the prior HNSCCUP.

Statistical analysis

Data were collated and curated in Excel and then imported into R for analysis. Statistical tests to generate *P* values depended on the type and distribution of the data and included the Wilcoxon rank sum, Pearson's χ^2 , and Fisher's exact test. A *P* value of <.05 was considered statistically significant.

Survival and recurrence were calculated from the end of radiation therapy to either an event or censorship. Survival curves for OS, primary emergence, and ipsilateral/CL neck recurrence were generated using the Kaplan-Meier (KM) method, with a log-rank test for significance. A regression model for OS was then built. Due to the low event rate, we were unable to perform multivariate analysis for primary emergence or CL neck recurrence. Ipsilateral neck recurrence was not analyzed further as these events represent intrinsic radioresistance rather than failure of the radiation therapy strategy.

The regression model was performed via logistic regression for initial univariate analysis, and then the Cox proportional hazards model (COXPH) for multivariate analysis, taking into account time to event data. Given the lack of statistical power due to low numbers of events, we also employed Bayesian inference via a Bayesian probabilistic regression model as a second means of validating our survival and regression findings. The models were adjusted for age, sex (as defined by the United Nations), Eastern Cooperative Oncology Group Performance Status (ECOG PS), tumor HPV status, smoking history, concurrent SACT, and nodal stage. These variables were selected based on their known prognostic value in HNSCCUP and based on their performance in the univariate analysis. We did not include

age and ECOG PS in the final Bayesian regression as they confounded the model due to small subgroup sizes.

The Bayesian regression was performed using the “brms,” “rstan,” and “Stan” packages in R.²⁹⁻³¹ Because of the lack of any robust research in this area, we elected to set a weakly informative prior for each variable and assumed a normal distribution. The prior was therefore set as normal (0, 2.5). This prior assumes a normal distribution, with the data falling within 2.5 standard deviations of the median. Full details regarding model parameters and performance are available in the supplemental figures E15 - E20 and Excel sheets “Bayesian_model_summary_OS,” “Bayesian_model_summary_Primary_Emergence” and “Bayesian_model_summary_CL”. All Rhat scores were <1.05, indicating good model convergence, with model performance also supporting the selection of the priors.

The Bayesian model gave a regression coefficient, which, when >0, favored the event and a probability of direction given as a percentage. A probability of 70% or greater was considered to give a moderate to high certainty.

Establishing acceptable differences in outcomes between INO and MUC

The risk of primary site emergence and potential reduced survival drives the philosophy behind comprehensive radiation with MUC in HNSCCUP. Knowing a priori, what, if any, increase in primary site emergence or reduction in OS with INO would be considered acceptable to the H&N cancer community was important to establish, allowing us to assess the impact of our findings.

To do this, we conducted a survey across the United Kingdom, including multiple known experts in the field, from both a radiation oncology and surgical oncology background. These clinicians were identified from earlier participation in a national consensus process on HNSCCUP.^{32,33}

The survey asked 2 questions:

- What do you think the acceptable OS difference should be when comparing INO versus MUC for HNSCCUP? Three options available: 0%-5%, 5%-10%, and 10%-15%
- What is an acceptable difference in rate of primary tumor emergence for HNSCCUP treated with INO versus MUC? Four options available: <5%, >5 & <10%, >10 & <15%, and 15%-20%

Results

Study population

Two hundred and sixty-nine patients were screened for study inclusion and 195 were eligible as shown in the Consort diagram in [Figure E1](#).

Table 1 Patient demographics

Variable	Whole cohort, N = 195		Radiation therapy technique		
	Data available (N)	N (%)	INO, N = 73 (%)	MUC, N = 122 (%)	P value*
Follow-up length (mo), median (IQR)	195	58 (42-72)	60 (43-70)	57 (41-73)	.86
Age at diagnosis, median (IQR)	170	59 (53-65)	59 (52-66)	59 (53-65)	.62
Sex	195				.70
Female		53 (27)	21 (28.8)	32 (26.2)	
Male		142 (73)	52 (71.2)	90 (73.8)	
ECOG PS	182				.20
0		138 (75.8)	52 (73.2)	86 (77.5)	
1		39 (21.4)	18 (25.4)	21 (18.9)	
2		5 (2.8)	1 (1.4)	4 (3.6)	
Smoking history	189				.77
Never smoker		53 (27)	22 (31.0)	31 (26.3)	
Smoker		136 (73)	49 (69.0)	87 (73.6)	
Alcohol history	166				.18
Drinks alcohol		138 (83)	53 (79.1)	85 (85.9)	
No alcohol		28 (17)	14 (20.9)	14 (14.1)	
HPV tumor DNA status	165				.010
Negative		56 (34)	29 (43.3)	27 (27.5)	
Positive		109 (66)	38 (56.7)	71 (72.5)	
Epstein arrVirus status	43				.019
Negative		42 (98)	22 (95.6)	20 (100)	
Positive [†]		1 (2.3)	1 (4.4)	0 (0)	
Nodal category	195				.78
N1/2a		120 (62)	43 (59)	77 (63)	
N2b		47 (24)	18 (24)	29 (24)	
N3a/b		28 (14)	12 (16)	16 (13)	

Abbreviations: ECOG PS = Eastern Cooperative Oncology Group Performance Status; HNSCCUP = head and neck squamous cell cancer; HPV = human papillomavirus; INO = involved neck only; MUC = mucosal.

* Wilcoxon rank sum test; Pearson's χ^2 test; Fisher's exact test.

[†] The EBV-positive case, treated with INO, was tested retrospectively and not at the time of initial diagnosis and treatment of HNSCCUP, see text for further details.

Demographics

Table 1 displays the demographics of the HNSCCUP study stratified by the 2 treatment categories. Seventy-three patients received INO radiation therapy (37.4%) and 122 patients received MUC radiation therapy (62.6%). Fifty-one of 73 patients treated with INO (69.9%) were from 2 centers and represented 89%-100% of the HNSCCUP population in those centers. Twelve of 18 centers used MUC in $\geq 80\%$ of their patients; 104/122 (85.2%) patients in MUC subgroup were from those centers.

The remaining 4 centers had a more even split between the 2 treatment strategies but contributed only 24 patients, 12.3% of the whole cohort. Details of the treatment strategy by center are shown in Table E1.

Table 2 shows the heterogeneity of treatment in the MUC population; 8 patients (6.6%) received bilateral neck radiation with no intentional MUC coverage, whereas 30 patients (24.6%) received ipsilateral neck and MUC radiation therapy but no CL neck treatment. Although 8 patients in the MUC category did not receive MUC radiation therapy, it is likely that by treating the neck bilaterally, potential MUC sites will have received an incidental elective dose of radiation.³⁴

There was no difference in age, sex, ECOG PS, smoking, or alcohol history between the 2 cohorts. In total, 72.5% of patients who underwent MUC radiation therapy had HPV-positive HNSCCUP versus 56.7% of patients in the INO radiation therapy group ($P = .010$). There was no difference in nodal staging between the treatment modalities.

Table 2 Treatment and toxicity details

Variable	Whole cohort		Radiation therapy strategy		P value*
	Data available (N)	N (%)	INO (N = 73), n (%)	MUC (N = 122), n (%)	
Primary radiation therapy	195	115 (59.0)	39 (53.4)	76 (62.3)	.22
Adjuvant radiation therapy	195	80 (41.0)	34 (46.6)	46 (37.7)	.22
Dose/fraction	195				<.001
55 Gy/20#		5 (2.6)166 (85.1)	4 (5.5)	1 (0.8)	
60-66 Gy/30-33#		24 (12.3)	69 (94.5)	97 (79.5)	
70 Gy/35#			0 (0)	24 (19.7)	
Concurrent SACT	195	128 (65.6)	43 (58.9)	85 (69.7)	.13
Drug	195				.35
Carboplatin		12 (6.2)	3 (4.1)	9 (7.4)	
Cetuximab		7 (3.6)	3 (4.1)	4 (3.3)	
Cisplatin		107 (55)	36 (49.3)	71 (58.2)	
Other agent		2 (1)	1 (1.4)	1 (0.8)	
N/A		67 (34.2)	30 (41.1)	37 (30.3)	
RT to involved neck and potential primary sites (PPS) only	195	30 (15.3)	0 (0)	30 (24.6)	NA
RT to involved neck and contralateral (CL) neck only (no PPS)	195	8 (4.1)	0 (0)	8 (6.6)	NA
RT to involved neck, PPS, and CL neck	195	84 (43.1)	0 (0)	84 (68.9)	NA
Median dose to PPS					
(Gy, IQR)	105	60 (54-65)	NA (NA)	60 (54-65)	NA
Median dose to CL neck (Gy, range)	89	54 (52-66)	0(0)	54 (52-66)	NA
Acute feeding tube insertion (%)	195	49 (25.1)	8 (11.0)	41 (33.6)	<.001
Long-term feeding tube Requirement (1 y or > since RT) (%)	195	7 (3.6)	0 (0)	7 (5.7)	.047
Mean dose to ipsilateral parotid (Gy), median (IQR)	164	40 (33-50)	40 (32-50)	40 (33-50)	.81
Mean dose to CL parotid (Gy), median (IQR)	182	11 (6-25)	6 (5-9)	24 (16-28)	<.001
Max dose to spinal cord (Gy), median (IQR)	190	39 (31-43)	31 (26-38)	41 (38-44)	<.001
Max dose to brainstem (Gy), median (IQR)	189	37 (24-44)	25(21-35)	42 (35-47)	<.001
NTCP for grade 2 xerostomia at 6 (mo)	182				<.001
Baseline 0, median (IQR)		28 (24-43)	24 (23-26)	42 (33-47)	<.001
Baseline 1, median (IQR)		45 (39-61)	39 (38-42)	59 (51-64)	

Abbreviations: INO = involved neck only; MUC = mucosal; RT = radiation therapy; SACT = systemic anticancer treatment. Bold values signifies statistically significant result.
 * Pearson's χ^2 test; Fisher's exact test; Wilcoxon rank sum test.

Dosimetry and toxicity

As shown in Table 2, MUC radiation therapy was associated with higher mean doses to OARs such as the CL parotid, spinal cord, and the brainstem ($P < .001$). Forty-one patients in the MUC radiation therapy cohort (34%) had

feeding tubes inserted in the acute setting, compared with 8 (11%) in the INO group ($P < .001$). Rates of feeding tube requirement at 12 months were also higher in the MUC radiation therapy cohort (5.7% vs 0%, $P = .046$). The NTCP model (Fig. 1) revealed a higher probability of moderate-severe xerostomia at 6 months for MUC radiation therapy

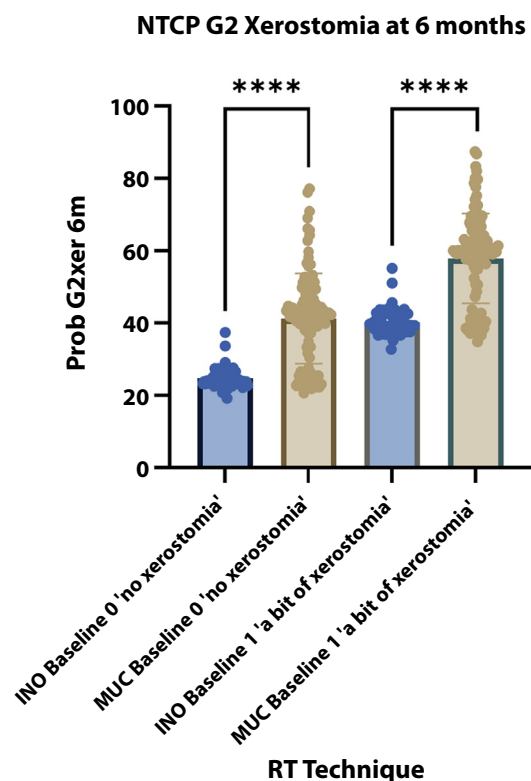


Fig. 1. Plot of Normal tissue complication probability (NTCP) for grade 2 xerostomia at 6 months in the presence and absence of baseline xerostomia. *Abbreviations:* INO = involved neck only; MUC = mucosal; RT = radiation therapy.

versus INO radiation therapy, in the presence of pre-existing xerostomia (median probability of 39% for INO radiation therapy vs 59% for MUC radiation therapy, $P < .001$) and in absence of pre-existing xerostomia (median

probability of 24% for INO radiation therapy vs 42% for MUC radiation therapy, $P < .001$). Graphical plots of the OAR dosimetry and feeding tube requirements are in the [Figures E2 and E3](#).

Disease-related outcomes and survival

The median duration of follow-up for the whole cohort was 58 months (IQR, 42-72 months).

[Table 3](#) displays the survival and recurrence outcomes for the patients. There were 3 instances of confirmed primary emergence, 2 in the INO radiation therapy (2.7%) cohort and 1 in the MUC radiation therapy cohort (0.8%). One of the INO radiation therapy primary emergencies was a nasopharyngeal emergence that was Epstein Barr Virus (EBV)-positive. EBV was not tested on the initial lymph node biopsy at HNSCCUP diagnosis, but was later confirmed retrospectively to have been positive. This case had HPV-negative disease. The MUC group's primary emergence had received an elective dose of radiation to the mucosa and a radical dose to the involved neck. All patients with primary emergence were surgically salvaged and disease free at time of analysis, with a mean follow-up time of 25 months.

In total, 7 patients had H&N cancers diagnosed following HNSCCUP treatment. The treating MDT felt that 4 were considered not to be primary site of emergence. Additional clinical details to justify these decisions were available for 3 cases. One had a new p16-negative oral tongue SCC; the original HNSCCUP presentation was p16-positive and included retropharyngeal node involvement, inconsistent with an oral cavity primary. A further patient developed a CL oral cavity SCC with associated nodal involvement in the CL neck 18 months later. A third case had p16-positive disease at presentation and p16-negative disease in a new

Table 3 Survival and recurrence

Variable	Whole cohort, N = 195		Radiation therapy technique		
	Data available N	N (%)	INO, N = 73 (%)	MUC, N = 122 (%)	P value*
Deceased	195	42 (21.5)	16 (21.9)	26 (21.3)	.92
Cause of death	195				.003
HNSCCUP		20 (10.3)	13 (17.8)	7 (5.7)	
Other causes		22 (11.3)	3 (4.1)	19 (15.6)	
N/A		153 (78.4)	57 (78.1)	96 (78.7)	
Primary emergence	195	3 (1.5)	2 (2.7)	1 (0.8)	.56
Regional recurrence	195	18 (9.2)	9 (12.3)	9 (7.4)	.25
Regional recurrence site	195				.13
Contralateral neck		5 (2.6)	4 (5.5)	1 (0.8)	
Ipsilateral neck		13 (6.7)	5 (6.9)	8 (6.6)	
Distant recurrence	195	18 (9.2)	11 (15.1)	7 (5.7)	.029

Abbreviations: HNSCCUP = head and neck squamous cell cancer; INO = involved neck only; MUC = mucosal.

* Pearson's χ^2 test; Fisher's exact test.

HNSCC. A summary of all subsequent cancers and whether they were defined as a true primary emergence can be found in [Table E2](#).

There were 5 instances of CL neck recurrence, with 4 in the INO radiation therapy cohort and 1 in the MUC radiation therapy cohort. Four were isolated CL neck recurrences; 1 also had distant metastases. Only 1 patient with an isolated CL neck recurrence died due to HNSCCUP. This patient was from the INO subgroup.

Five-year OS was 80% (95% CI 70%-90%) for INO radiation therapy and 82% (95% CI 75%-90%) for MUC radiation therapy, $P = .74$, with 96% (95% CI, 91%-100%) freedom from primary emergence at 5 years for INO radiation therapy and 99% (95% CI, 97%-100%) for MUC radiation therapy, $P = .31$. CL neck recurrence was higher for INO radiation therapy ($P = .044$), with 5 year rates of CL neck recurrence-free survival (RFS) 94% (95% CI, 88%-100%) in the INO group versus 99% (95% CI, 97%-100%) for MUC radiation therapy.

The KM survival curves for OS, primary emergence, CL neck recurrence, HNSCCUP associated survival, and other causes of mortality are displayed in [Figure 2](#). [Figure 2](#) also displays the stratified KM survival curve generated from the multivariate regression analysis. The KM survival curves (and survival, in [Tables E3-E5](#)) show negligible differences in primary emergence and OS.

There were 42 patients (21%) who had died by the time of analysis, with 78% of the INO radiation therapy and 79% of the MUC radiation therapy group still alive. Distant recurrence was higher in the INO radiation therapy cohort (15% vs 5.7%, $P = .029$), death from HNSCCUP was also higher for INO radiation therapy (18% vs 5.7%, $P = .003$) with metastatic disease being the main cause. Death from other causes was higher in the MUC radiation therapy cohort (16% vs 4.1%, $P = .003$).

Subgroup analysis for the 109 patient HPV-positive cohort, stratified by treatment strategy, was carried out. KM survival curves are available in [Figures E4-E8](#). CL neck RFS at 5 years was no different between INO 100% (95% CI 100%-100%) and MUC 99% (95% CI 96%-100%) ($P = .46$). There was no difference in freedom from primary emergence at 5 years, INO 100% (95% CI, 100%-100%) and MUC 100% (95% CI, 100%-100%) ($P = 1$). HNSCCUP-specific survival was also no different between INO 97% (95% CI, 90%-100%) and MUC 96% (95% CI, 90%-100%) ($P = .62$). There was a trend for lower 5-year OS with MUC 84% (95% CI, 75%-94%) versus 97% with INO (95% CI, 90%-100%), but this did not reach statistical significance ($P = .078$). This was driven by non-HNSCCUP death, with non-HNSCCUP survival lower at 5 years, with MUC 88% (95% CI 79%-97%) compared to 100% with INO (95% CI, 100%-100%) ($P = .016$).

The HPV-negative cohort consists of 56 patients, 29 treated with MUC and 27 with INO. Small numbers when stratified by treatment prevent meaningful subgroup analysis, but this was carried out for completeness. KM survival curves are available in [Figures E9-E13](#). Comparison of

demographics in [Table E6](#) between the HPV-positive and -negative subgroups shows that HPV-negative disease is associated with older patients, poorer PS, a smoking history, and increased nodal burden. Comparison of outcomes between HPV-positive and -negative disease ([Table E6](#)) showed that while regional recurrence rates were higher in HPV-negative disease [12 of 56 patients with HPV-negative disease (21%) versus 3 of 109 (3%) with HPV-positive disease $P < .001$], 75% of the HPV-negative regional recurrences were in the ipsilateral neck. Distant recurrence was also more likely in the HPV-negative subgroup, with 10/56 (18%) experiencing distal metastases versus 5 of 109 (5%) in the HPV-positive group ($P = .005$).

CL neck RFS at 5 years was no different between INO 88% (95% CI, 75%-100%) and MUC 100% (95% CI, 100%-100%) ($P = .081$). There was no difference in freedom from primary emergence at 5 years, INO 96% (90% CI 100%-100%), and MUC 96% (95% CI 89%-100%) ($P = .99$). HNSCCUP-specific survival was poorer with INO 65% (95% CI 49%-87%) and MUC 100% (95% CI 100%-100%) ($P = .043$). Although OS was also numerically lower with INO 62% (95% CI, 45%-84%) versus MUC 81% (95% CI, 67%-98%), this did not reach statistical significance ($P = .078$). Non-HNSCCUP survival in the HPV-negative population was no different at 5 years, with MUC 81% (95% CI, 67%-98%) compared to 95% with INO (95% CI, 85%-100%) ($P = .25$).

Standard and Bayesian regression analysis

The univariate logistic, multivariate COXPH, and Bayesian models are displayed in [Table 4](#), with the multivariate COXPH forest plot in [Figure E14](#). The only independent predictors of OS were nodal stage and concurrent SACT. MUC radiation therapy was associated with poorer OS, but this was nonsignificant (hazard ratios = 1.40, 0.67-2.93, $P = .38$). However, Bayesian regression supported this finding with a probability of 69%, [Figures E15 and E16](#). Crucially, INO radiation therapy was not associated with poorer OS on either univariate, multivariate, or Bayesian approaches.

Bayesian regression was used to assess CL neck recurrence and primary emergence ([Table E7](#) and [Figures E17-E20](#)). MUC radiation therapy was associated with a lower risk of primary emergence and CL neck recurrence. Nodal stage and HPV status were also found to influence OS, CL neck recurrence, and primary emergence.

Acceptable differences in outcomes between INO and MUC

We received 32 responses to the survey of national experts. Eighty-four percent of responders deemed a reduction in OS of up to 5% between the 2 treatment approaches to be acceptable. For primary site emergence, 53% of respondents suggested that a difference of up to 5% between radiation therapy strategies was acceptable, whereas 28% deemed a

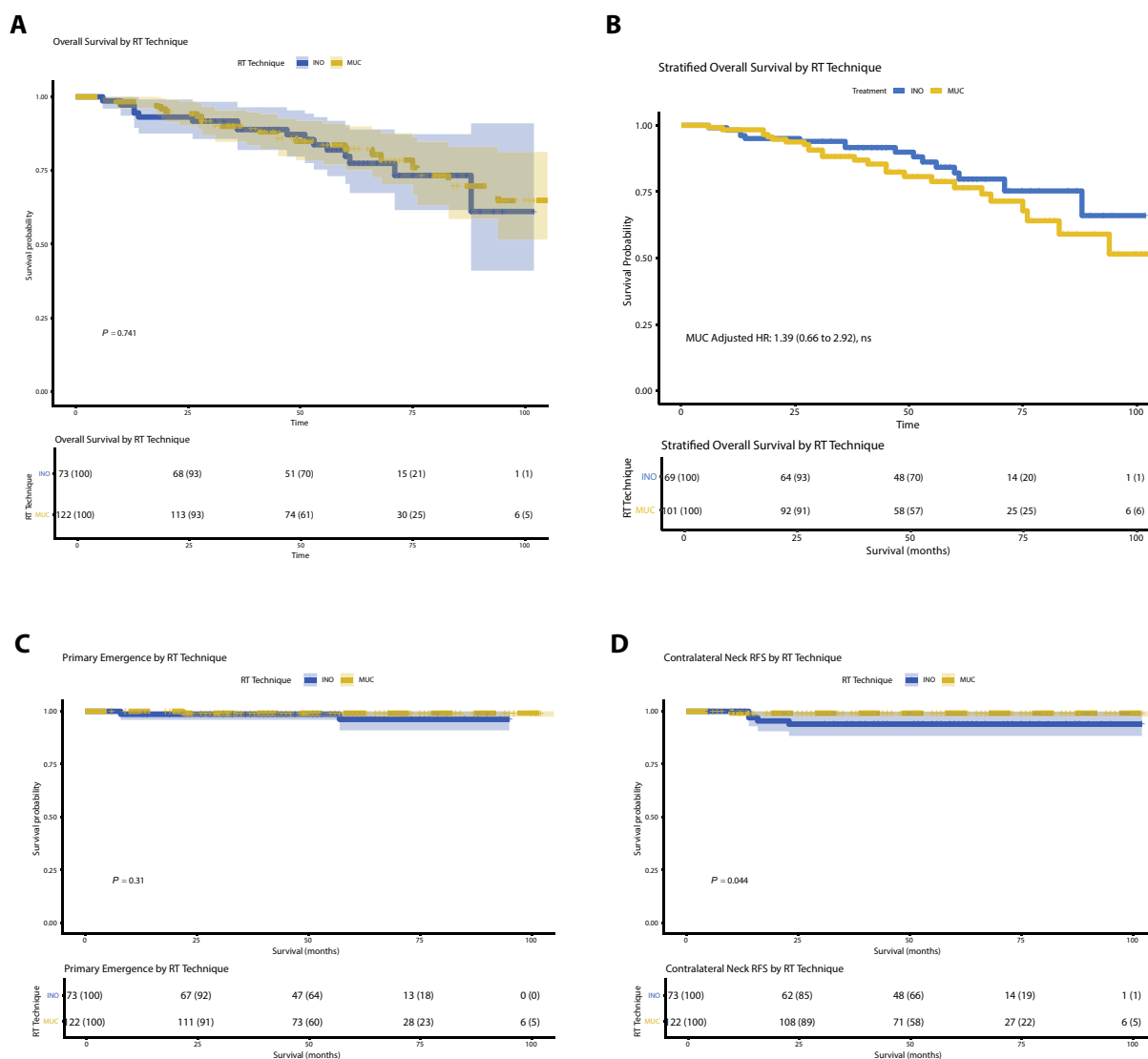


Fig. 2. (A) Kaplan-Meier overall survival curve by RT technique. *Abbreviations:* INO = involved neck only; MUC = mucosal; RT = radiation therapy. (B) Kaplan-Meier stratified overall survival by RT technique. *Abbreviations:* HR = hazard ratio; INO = involved neck only; MUC = mucosal; RT = radiation therapy. (C) Kaplan-Meier survival curve for primary site emergence by RT technique. *Abbreviations:* INO = involved neck only; MUC = mucosal; RT = radiation therapy. (D) Kaplan-Meier survival curve for contralateral neck recurrence by RT technique. *Abbreviations:* INO = involved neck only; MUC = mucosal; RFS = recurrence-free survival; RT = radiation therapy. (E) Kaplan-Meier survival curve for HNSCCUP-specific survival by RT technique. *Abbreviations:* INO = involved neck only; MUC = mucosal; RT = radiation therapy. (F) Kaplan-Meier survival curve for other cause mortality by RT technique. *Abbreviations:* INO = involved neck only; MUC = mucosal; RT = radiation therapy.

difference of up to 10% acceptable. See [Figures E21 and E22](#) for more detail.

Discussion

We present the largest series to date of patients with unilateral HNSCCUP treated radically with radiation therapy in the HPV and FDG PET-CT era.^{13,35} Furthermore, 37% of

our cohort received INO radiation therapy, allowing a comparison of this and the historical standard strategy of including the CL neck and pharyngeal mucosa in the elective treatment volume. Key findings are acceptably low rates of primary site emergence and CL neck failure in the INO group, with an increased rate of toxicity in the MUC cohort. Although there was no difference in OS between the treatment categories, those undergoing INO radiation therapy were more likely to die from HNSCC and those receiving

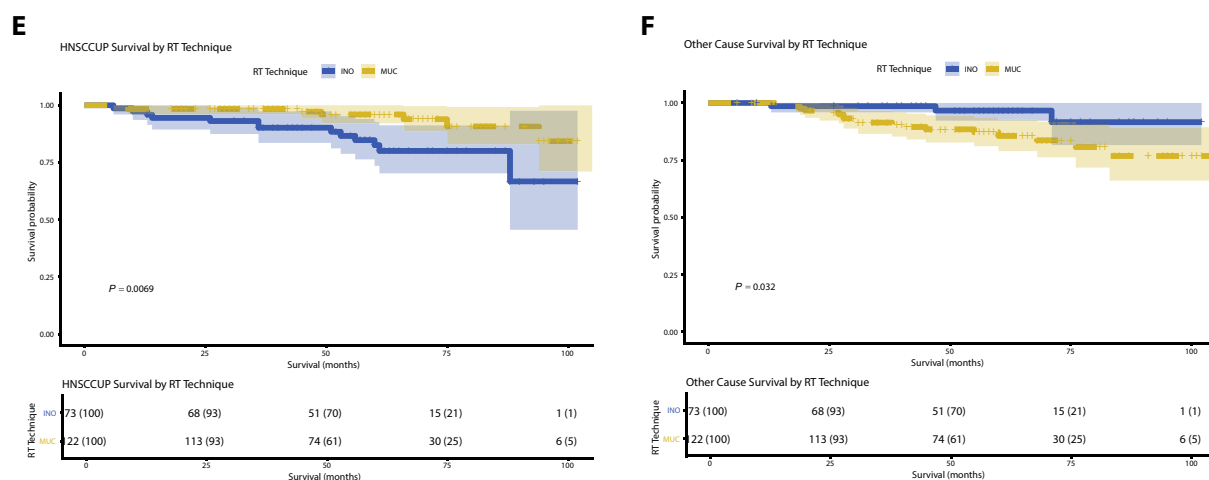


Fig. 2. Continued.

MUC radiation therapy were more likely to die from a non-HNSCC cause.

Our cohort reflects contemporary epidemiology and diagnostics. Sixty-six percent of patients had HPV-positive disease, and all underwent FDG PET-CT during their diagnostic workup. The proportion of HPV-positive disease is higher than that previously reported for primary-detected oropharyngeal squamous cell cancer (SCC) in the UK, potentially reflecting the later time period under evaluation in the current analysis³⁶ It is also higher than recently reported in a phase IV study from The Danish Head and Neck cancer Study Group (DAHANCA), where 43% of cases of HNSCCUP had HPV-driven disease. They issue a timely reminder that not all HNSCCUP have undetected oropharyngeal SCC.³⁷ This consideration remains relevant to our population as baseline EBV testing may have avoided the emergence of a primary in 1 patient treated in the INO group with HPV-negative disease, where the presenting lymph node was retrospectively found to be EBV-positive after the emergence of an EBV-positive nasopharyngeal primary. This supports EBV testing of HPV-negative HNSCCUP, which is known to improve diagnostic accuracy.³⁸

Stratifying by tumor HPV status demonstrates no difference in OS, CL neck RFS, and freedom from primary emergence in the HPV-positive and -negative subgroups, regardless of radiation therapy strategy. For the HPV-positive cohort, HNSCCUP-specific survival was no different between INO and MUC, whereas non-HNSCCUP survival was better with INO. In the HPV-negative group, although HNSCCUP-specific survival was poorer with INO than MUC, this appeared to be driven by ipsilateral neck and distal relapse. More extensive irradiation with MUC is unlikely to modify this pattern of disease relapse, which is related to tumor biology rather than radiation therapy strategy. These observations from subgroup analysis come with the caveat

of small numbers, especially in the HPV-negative group, and with the knowledge that confounders affecting outcomes (age, PS, smoking history, and nodal burden) were less favorable in the HPV-negative group.

As expected, INO radiation therapy results in lower doses to OARs with significant and clinically meaningful differences recorded.²⁷ Given the retrospective and multicenter nature of the work, we focused on OARs that we felt would be delineated consistently across all sites. The inclusion of other OARs, such as esophagus or swallowing OARs, was desirable but may have introduced variations in practice and confounded results. Equally, evaluation of toxicity retrospectively from routinely collected clinical data is notoriously difficult. We therefore have focused on a robust single variable, the presence of an enteral feeding tube, which is most likely to have been recorded reliably. It is acknowledged that the presence of a feeding tube in the treatment phase includes both those placed prophylactically and reactively and is therefore not a direct surrogate for toxicity. However, prophylactic gastrostomies are placed when it is anticipated that they will be required. Therefore, the presence of an enteral feeding tube seems a reasonable, if crude, indicator of likely acute toxicity. In the acute, and more importantly, late setting, the proportion of patients with enteral feeding tubes was significantly lower with INO radiation therapy with no patients requiring long-term enteral feeding in this group, compared to 5.7% in MUC. The same pattern is likely to be true with xerostomia when we consider the significant difference in dose received by the CL parotid. Based on the Beetz NTCP model,²⁷ the probability of patients having moderate-to-severe xerostomia 6 months posttreatment (assuming no baseline symptoms) was 18% higher in the MUC cohort. This is in keeping with prior reports from single-center series.^{39,40}

Determining if a subsequent cancer is emergence of the unknown primary at the index site, or a second cancer, is

Table 4 Univariate, multivariate and Bayesian regression for overall survival

Characteristic	Univariate regression				Multivariate regression				Bayesian regression Coefficient/Probability of direction%
	N	OR	95% CI	P value	N	HR	95% CI	P value	
Age at diagnosis	170	1.08	1.04 to 1.13	<.001	170	1.04	0.99 to 1.09	.087	N/A
ECOG PS	195				170				N/A
0		—	—			—	—		
1		3.88	1.75 to 8.57	<.001		1.50	0.70 to 3.22	.29	
2		3.71	0.47 to 23.7	.16		1.46	0.30 to 7.11	.64	
Unknown		1.67	0.35 to 6.02	.46		2.67	0.73 to 9.81	.14	
Sex	195				170				
Female		—	—			—	—		—
Male		1.25	0.58 to 2.88	.58		0.92	0.42 to 2.00	.83	0.13, 60
Smoking history	195				170				
Never smoker		—	—			—	—		—
Smoker		2.02	0.87 to 5.29	.12		0.75	0.30 to 1.85	.53	0.24, 68
Unknown		6.57	1.04 to 42.5	.039		2.18	0.49 to 9.74	.31	2.11, 98
Nodal category	195				170				
N1/2a		—	—			—	—		—
N2b		2.76	1.21 to 6.27	.015		2.17	0.71 to 6.67	.17	0.77, 91.5
N3a/b		4.88	1.94 to 12.3	<.001		3.11	1.15 to 8.37	.025	1.28, 98.6
Tumor HPV status	195				170				
Negative		—	—			—	—		—
Positive		0.29	0.13 to 0.62	.002		0.92	0.32 to 2.58	.87	−0.54, 82.1
Unknown		0.55	0.19 to 1.45	.24		2.45	0.73 to 8.17	.14	−0.58, 83.4
Treatment	195				170				
INO		—	—			—	—		—
MUC		0.96	0.48 to 1.98	.92		1.39	0.66 to 2.92	.38	0.20, 69
Concurrent SACT	195				170				
No		—	—			—	—		—
Yes		0.38	0.19 to 0.78	.006		0.40	0.17 to 0.95	.038	−0.64, 93.6
Abbreviations: ECOG PS = Eastern Cooperative Oncology Group Performance Status; HPV = human papillomavirus; HR = hazard ratio; OR = odds ratio; SACT = systemic anticancer treatment.									

challenging in daily clinical practice and also in this study. Anonymized clinical documentation and opinion from the local treating team, histology, and radiology reports were reviewed centrally to evaluate subsequent cancer diagnoses and a judgment was made. It is possible that the rate of local failure is underestimated by us, deeming 4 subsequent cancers to be second, unrelated cancers. However, these cases were split equally between the INO and MUC subgroups, therefore not altering the overall pattern demonstrated.

The extent of MUC irradiation in the MUC subgroup was not further categorized. Several approaches to this exist, ranging from comprehensive volumes including all possible sites of disease (nasopharynx, oropharynx, larynx, and

hypopharynx) to coverage of specific ipsilateral subsites only.^{41–43} Our MUC subgroup is likely to have included a variety of MUC target volumes, but again given the multicenter and retrospective nature of the work, it was felt that 2 clearly defined treatment categories of INO and MUC radiation therapy were more likely to be identified robustly. Further stratification of the MUC cohort by extent of MUC irradiation is unlikely to have affected the pattern of results we report. The crucial question of the rate of primary site emergence with INO radiation therapy is not affected by the extent of MUC irradiation in the MUC subgroup comparator. The rate of toxicity in the MUC cohort may be over or underestimated depending on the proportion of entire

pharyngeal axis volumes versus ipsilateral specific subsites. Overall, it is likely that the lower dose to OARs and lower rate of enteral feeding with INO is representative of the broad treatment strategies, and is in keeping with data in oropharyngeal SCC examining unilateral versus bilateral radiation therapy.⁴⁴

No difference in the proportions of deaths is demonstrated between INO and MUC cohorts, with 21% of INO subgroup deceased and 22% of MUC ($P = .96$). This is in keeping with KM survival analysis, when stratified by treatment strategy there was no difference in 5-year OS (80% for INO radiation therapy, 82% for MUC radiation therapy, $P = .84$). These 5-year OS rates compare favorably with the recent work from DAHANCA in HNSCCUP, with 5-year OS of 73% reported. The difference may be explained by the higher proportion of HPV-positive disease in the current study, as described earlier, although it is interesting to note that on multivariate analysis tumor HPV status had little impact on OS in this work, which is out of keeping with prior reports.¹⁹

Factors affecting OS were examined using univariate and multivariate analysis. The only independent predictor of OS was nodal stage; a nonsignificant association with poorer OS was seen with MUC radiation therapy. INO radiation therapy was not associated with poorer OS on univariate or multivariate approaches. The lack of difference in OS between INO and MUC subgroups in all these analyses was despite a lower proportion of HPV-positive disease in the INO group, although as noted above, tumor HPV status had little impact on multivariate OS analysis. The INO group also tended to have a lower proportion of N1/2a disease and a higher proportion of N3 disease, although these differences were not statistically significant.

HNSCCUP-specific death was higher in the INO group, driven mainly by an increased rate of distant metastases. Thirteen of 74 patients who received INO radiation therapy died of HNSCCUP, with 11 of those having distant metastases. Reasons for this are speculative, but in general, factors associated with distant metastasis in HNSCC are nodal burden, extranodal extension, locoregional residual disease, and HPV-negative status, all of which may play a part here.⁴⁵ In particular, the INO cohort had a higher proportion of HPV-negative disease, and although not statistically significant, numerically lower rates of concurrent SACT. We also observed that the INO group was numerically (but not statistically) more likely to have had up front neck dissection than the MUC cohort (46.6% vs 37.7%). It seems that the combined modality treatment package in the INO group tended to be surgery plus adjuvant radiation therapy, whereas it was more likely to be chemoradiation in the MUC group. Rates of CL neck failure were also higher with INO, and while the majority were successfully salvaged, one could hypothesize that active regional disease allowed the dissemination of distant metastases. Regardless of etiology, it is a finding that merits further investigation. Out with H&N cancer, this study lacks disease-specific cause of death

data. Therefore, it is not possible to infer reasons why non-HNSCCUP death rates are higher in the MUC radiation therapy cohort. However, it is plausible to infer that treatment morbidity may be a factor, and as such would be an important topic for future research. A recent study from Princess Margaret Cancer Center, Toronto, reports on radiation late toxicities (RLT) in a cohort of 7622 patients who underwent radiation therapy for H&N cancer.⁴⁶ In total, 12.6% developed RLT in the absence of disease progression. Higher volume of neck irradiated was associated with increased risk of RLT, which is relevant to the INO versus MUC comparison, given that 75% of the MUC group had bilateral neck irradiation versus 0% of the INO cohort. Importantly, they found that patients who develop RLTs experience worse survival (hazard ratio = 2.1, 95% CI, 1.8-2.5, $P < .001$).

The rate of primary site emergence was no different between the treatment groups: 2.7% with INO radiation therapy and 0.8% with MUC. This pattern is confirmed with KM survival analysis, with 96% freedom from primary emergence seen with INO and 99% with MUC. Whichever way assessed, primary site emergence is low with INO radiation therapy, questioning the convention of delivering an elective dose of radiation therapy to potential MUC primary sites. It is generally accepted that elective treatment of the neck is indicated, and the associated morbidity justified, when the risk of micrometastatic disease is 15%-20%.⁴⁷ It would seem reasonable that similar principles, or potentially with a higher threshold given the higher morbidity of treatment, might be applied to elective MUC treatment. HNSCCUP diagnosed in the contemporary era does not meet this threshold for elective treatment, whether that be of the mucosa or CL neck.

The levels of involved lymph nodes were not collected for the purposes of this study but are likely to be similar to those previously described by the ENT INTEGRATE work from which the majority of this sample was drawn. They report that the nodal level with the greatest nodal burden at presentation was level II, affecting 74.9% of their cohort.¹⁹ This is consistent with the recent work from DAHANCA where level II was also the most frequently involved nodal level.³⁵ Prior dosimetric work found that an incidental dose of at least 40 Gy was received by the oropharyngeal mucosa with INO. The majority of patients (80%) in that study had level 2 involvement.³⁴ It would seem reasonable to postulate that the excellent local control demonstrated with INO may be because of incidental eradication of microscopic disease in potential primary sites. The observation that poorer local control (ie, increased rates of primary site emergence) is reported with neck dissection alone compared to the inclusion of radiation therapy in the treatment of HNSCCUP would also be consistent with this hypothesis. Given that over half of our INO cohort received concurrent chemotherapy this may have also contributed to the eradication of microscopic disease.

Responses to our anonymized multidisciplinary survey further allow us to contextualize reported outcomes.

The majority of responders felt that a reduction of up to 5% in OS and an increase of up to 5% in the rate of primary site emergence with INO compared to MUC radiation therapy would be acceptable. The survey did not include patient opinion, which, of course, is crucial to discussing the trade-off between morbidity of treatment and cancer control.⁴⁸ However, the lack of difference in both OS and primary site emergence between the treatments is likely to be highly reassuring to both clinicians and patients.

The limitations of our study are as acknowledged earlier. As is often the case with retrospective data, toxicity results are limited. Case evaluation relied on identification by local centers, either as part of the ENT INTEGRATE work or at the inception of this project. Selection bias may therefore confound results, although screening with inclusion and exclusion criteria may minimize this. The nonrandomized use of INO or MUC may mean the cohorts have additional imbalances beyond those identified, with patient and tumor characteristics potentially impacting treatment choice. However, the majority of patients in the study (88%) were managed in centers with a clear preference toward either INO or MUC, suggesting clinician bias rather than patient/tumor characteristics was the main factor in the choice of radiation therapy strategy. Existing clinician bias is also in keeping with poor recruitment to the previously described prospective study, and the failure to generate randomized evidence in this space. One center treated 50 of 73 (68.5%) of patients receiving INO, which should be borne in mind when considering the generalizability of the results. Diagnostic oropharyngeal surgery was heterogeneous in this cohort, as previously described in the ENT INTEGRATE work.¹⁹ Although more extensive sampling has been shown to identify more primary tumors, the impact on survival is not yet established. If anything, we might expect more extensive diagnostic surgery, eg, tongue base mucosectomy, to result in even less primary site emergence. Given that less than 20% of patients underwent tongue base mucosectomy during their diagnostic workup, the local failure rate with INO in combination with this emerging technique may be even lower.

Finally, the broad categorization of the treatment strategies into INO and either a CL neck or a prophylactic MUC volume limits the opportunity to evaluate different MUC irradiation strategies, but it was felt that clarity on this categorization was more important than further subgroup stratification within the MUC approach.

Conclusions

Acceptably low rates of primary site emergence, lower toxicity, and no difference in OS when compared with prophylactic MUC radiation therapy suggest that INO radiation therapy in HNSCCUP may be a feasible alternative in contemporary practice.

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